TEMPLATE LaTeX Lab Report 2025
(Replace with name of experiment)

Physics 36X Laboratory ##

DATE September 22, 2025
TA:
Group number:

Names of everyone in the lab group
One person per line

The University of Calgary



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1 Objective

State the purpose of the experiment and the goals you want to accomplish. Use In 2-3 sentences with strong verbs. Use active voice instead of passive voice (forms of the verb 'to be' indicate passive voice):

- We determined .... Instead of: ... was determined.
- We calculated .... Instead of: ... was calculated.
- We compared .... Instead of: ... was compared.
- We investigated .... Instead of: ... was investigated.

2 Introduction and Theory

Include a concise summary of the relevant theory. Cite the equations you use with proper references such as the textbook or laboratory write up you worked from.

3 Experimental Methods

Describe the steps your group took to conduct the experiment and the needed materials and equipment. Provide sketches and figures of the experimental setup with appropriate captions.

4 Data

Present your organized data in this section. These data (the word 'data' is plural) must match the raw data shown to your TA before exiting the lab.

Table 1: Caption describing the data contained in the table

Column 1	Column 2	Column 3
Data Row 1, Col 1	Data Row 1, Col 2	Data Row 1, Col 3
Data Row 2, Col 1	Data Row 2, Col 2	Data Row 2, Col 3

Table 2: Caption describing the data contained in the table

	Measurement	Units
Row 1 measurement	Number	unit
Row 2 measurement	Number	unit

If you have a large table already made in Excel you can easily convert it to latex using `https://www.tablesgenerator.com/`, this can even add the table boarders!

5 Data Analysis

Data analysis presents the calculations and thinking about the numerical results of the laboratory work. Display any and all needed calculations, including sample uncertainty calculations. This section frequently has additional tables (comparisons, graphing tables, slope/intercept, etc.), and graphs.

To have a centered sub-heading to separate sections in this part do as follows:

Math Syntax

Math can either be done inline, $5 + 5 = 10$, or centered,

$$a^5 a^6 = a^{11}$$

or with enumeration

$$\boxed{\prod_{i=1}^{n} x^i = x^{\sum_{i=1}^{n} i} = x^{n(n+1)/2}} \tag{1}$$

which can be referenced later using "\ref{eq:eqreference}", 1. If you have an important value to display you can also box it, as above.

To have a sequence of equations strung vertically together use the following:

$$\begin{aligned} a &= b \\ &= c \\ &= d \end{aligned}$$

Figures

A figure can be inserted as follows:



Figure 1: Figure caption describing any symbols and equations on the figure, as well as what is being plotted if it is a plot.[1]

Figure captions appear below figures, but Table captions appear above. I have no idea why.

6 Discussion

Compare your obtained values to the expected ones using the t-test. Comment on the agreement and possible sources of errors (random and systematic). Make sure that the errors are reasonable given your results.

7 Conclusion

Report values again and comment on their agreement with accepted values. Provide any concluding thoughts. For this class, this section should be short.

References

[1] Wikipedia (2022) "Andromeda Galaxy," https://en.wikipedia.org/wiki/Andromeda_Galaxy#/media/File:Andromeda_Galaxy_560mm_FL.jpg

[2] Christopher Cully (2018) "Measurement Uncertainty," University of Calgary, Department of Physics and Astronomy